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Large-area heater

Abstract

An embodiment of the present invention provides a large-area heater. The large-area heater comprises: a heating plate including a central area in which heat is concentrated and a peripheral area surrounding the central area; and a plurality of unit heaters for heating at least a part of the central area and at least a part of the peripheral area, wherein each of the plurality of unit heaters comprises: a plurality of heat emitting areas producing different amounts of heat; a plurality of power wires for independently transmitting power to the plurality of heating areas, respectively; and pairs of power terminals which are arranged at the corner of each of the unit heaters so as to supply power to the plurality of power wires, the number of pairs of power terminals corresponding to the number of the plurality of heat emitting areas.

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Background/Summary

FIELD
(1) The present invention relates to a heater for semiconductor manufacturing equipment.
BACKGROUND
(2) A heater converts electrical energy into thermal energy. Typically, an oven for manufacturing

semiconductors equipped with a heater is essential for the process of removing air bubbles from the adhesion between the glass substrate and the lower substrate in process of manufacturing a flexible OLED module, and is an important facility for managing the final yield of the OLED module. The heater is a main part in semiconductor equipment for controlling heat source, and it is required to enlarge heater modules due to the enlargement of OLED displays. When the heater module is enlarged, the difficulty of heat source control is reduced, and in particular, there is an effect of reducing the cost of parts.
(3) However, when the heater module is enlarged, the thermal uniformity is lowered, and a heater pattern design and module assembly technology are required according to the area expansion. In particular, due to the combination of characteristics that semiconductor processes have many thermal processes and thermal process makes it difficult to control heat sources as the area increases, it becomes difficult to respond to equipment for manufacturing OLED display.
SUMMARY
(4) It is intended to provide a large-area heater having a structure that is relatively easy to control a heat source.
(5) There is provided a large-area heater. The large-area heater may include a heating plate having a central region to which heat is concentrated and a peripheral region surrounding the central region, and a plurality of unit heaters configured for heating at least a portion of the central region and at least a portion of the peripheral region. The plurality of unit heaters may include a plurality of heating regions configured for generating different amounts of heat, a plurality of power lines configured for independently transmitting power to each of the plurality of heating regions, and a plurality of power connection terminal pairs disposed at a corner of the unit heater to supply power to the plurality of power lines, wherein the number of the plurality of the power connection terminal pairs corresponds to the number of the plurality of heating regions.
(6) In one embodiment, the heating region may include a single zone and a multi-zone having a plurality of sub-zones generating different amounts of heat. The plurality of sub-zones may be supplied with power through one power line.
(7) In one embodiment, the plurality of unit heaters comprises left, right and vertically symmetrical heating regions.
(8) In one embodiment, the unit heater may include a first heating region corresponding to the central region and a second heating region corresponding to the peripheral region. A pattern density of a heater pattern disposed in the first heating region may be smaller than a pattern density of a heater pattern disposed in the second heating region.
(9) In one embodiment, the unit heater may include a first heating region corresponding to the central region and a second heating region corresponding to the peripheral region. A cross-sectional thickness of a heater pattern disposed in the first heating region may be different from a cross-sectional thickness of a heater pattern disposed in the second heating region.
(10) In one embodiment, the heating region may include a heater pattern formed by electrically connecting a plurality of sub-patterns.
(11) In one embodiment, the plurality of sub-patterns may include a first connector and a second connector, both having slit formed thereon and a sub heating element bent in a zigzag shape to have a plurality of bending points between the first connector and the second connector.
(12) In one embodiment, a temperature deviation of the central region is adjusted by controlling a temperature of the second heating region corresponding to the peripheral region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Hereinafter, embodiments of the invention will be described with reference to the

accompanying drawings. For the purpose of easy understanding of the invention, the same elements will be referred to by the same reference signs. Configurations illustrated in the drawings are examples for describing the invention, and do not restrict the scope of the invention.
(2) FIG. **1** exemplarily illustrates a unit heater constituting a large-area heater;
(3) FIG. **2** exemplarily illustrates a large-area heater consisting of four unit heaters;
(4) FIG. **3** exemplarily illustrates coupling sub-patterns to form a heating region;
(5) FIG. **4** exemplarily illustrates a process for manufacturing a large-area heater;
(6) FIG. **5** exemplarily illustrates one embodiment for implementing a different amount of heat generated for each heating region; and
(7) FIG. **6** exemplarily illustrates another embodiment for implementing a different amount of heat generated for each heating region.

DETAILED DESCRIPTION

(8) Embodiments which will be described below with reference to the accompanying drawings can be implemented singly or in combination with other embodiments. But this is not intended to limit the present invention to a certain embodiment, and it should be understood that all changes, modifications, equivalents or replacements within the spirits and scope of the present invention are included.
(9) Terms such as first, second, etc., may be used to refer to various elements, but, these element should not be limited due to these terms. These terms will be used to distinguish one element from another element.
(10) The terms used in the following description are intended to merely describe specific embodiments, but not intended to limit the invention. An expression of the singular number includes an expression of the plural number, so long as it is clearly read differently. The terms such as “include” and “have” are intended to indicate that features, numbers, steps, operations, elements, components, or combinations thereof used in the following description exist and it should thus be understood that the possibility of existence or addition of one or more other different features, numbers, steps, operations, elements, components, or combinations thereof is not excluded.
(11) FIG. **1** exemplarily illustrates a unit heater constituting a large-area heater.
(12) One large-area heater **100** (see FIG. **2**) includes a plurality of unit heaters **10**. The unit heater **10** includes a plurality of heater patterns attached to an insulating plate having a width L.sub.1 and a length L.sub.2. In one embodiment, the insulating plate may be a mica plate, but is not limited thereto. The insulating plate may include an upper insulating plate and a lower insulating plate, and a plurality of heater patterns may be interposed between the upper insulating plate and the lower insulating plate.
(13) The unit heater **10** may include a plurality of heating regions A, B, C, and D. The heating region may be single zones C, D or multi-zones A, B. The single zone and the multi-zone may be formed by a plurality of heater patterns. The multi-zone may include a plurality of sub-zones, for example, a**1**, a**2**, a**3**, a**4** in the heating region A, which receive power supplied through one power line **12**. The heater pattern may be formed by coupling a plurality of sub-patterns. Each of the plurality of heating regions A, B, C, and D may generate a different amount of heat. The plurality of heating regions A, B, C, and D may be disposed to be spaced apart from each other by a certain interval for electrical and thermal insulation. On the other hand, a temperature sensor may be disposed for each heating region. A method of manufacturing each of the plurality of heating regions A, B, C, and D have different amounts of heat will be described in detail below with reference to FIGS. **5** and **6**.
(14) Each of the plurality of heating regions A, B, C, and D is powered by a single power supply. The power connection region **11** is located near one edge of the unit heater **10**. In the power connection region **11**, the same number of power connection terminal pairs as the number of heating regions are disposed. In the case of the unit heater **10** illustrated in FIG. **1**, since there are four heating regions, four power connection terminal pairs are disposed in the power connection

region **11**. The power line **12** is disposed for each heating region without distinction between single zones and multi-zones. One end of the power line **12** is electrically coupled to the power connection region **11**, and the other end is electrically coupled to the heating regions A, B, C, and D. That is, the power line may extend along the circumference of the unit heater **10** from the pair of power connection terminals. The power line **12** may be formed of the same material as the heater pattern, and therefore generate heat by the transmitted power.
(15) Unlike single zones C and D, since each multi-zones A and B consist of a plurality of sub-zones, power supplied through the power line **12** is distributed to respective sub-zones within each multi-zone. In one embodiment, as illustrated in FIG. **1**, each sub-zone may be coupled in series. In another embodiment, each sub-zone may be coupled to a sub power line (not shown) branched from the other end of the power line **12**.
(16) In the unit heater **10**, the plurality of heating regions A, B, C, and D may be disposed in consideration of the temperature uniformity of the large-area heater **100** in which the plurality of unit heaters **10** are coupled. Compared to the case where the unit heater **10** is implemented with the same heater pattern (that is, the amounts of heat generated in all heating regions are the same), it is very difficult to control the temperature deviation that may occur on the unit heater **10** as well as the temperature deviation that occurs on the large area heater **100**. On the other hand, in the structure of individually controlling each of the plurality of heating regions (that is, power control for each single zone and sub-zone), it is difficult to arrange the power line for transmitting power, and since it is necessary to provide a separate temperature controller for each zone, the manufacturing cost increases. The arrangement of the heating regions in consideration of temperature deviation control will be described in detail with reference to the large-area heater of FIG. **2**.
(17) FIG. **2** exemplarily illustrates a large-area heater consisting of four unit heaters.
(18) Referring to FIG. **2**, the large-area heater **100** may include four unit heaters **10***a*, **10***b*, **10***c*, and **10***d* and a heating plate (not shown) having a size of having a width 2L.sub.1 and length 2L.sub.2. Each of the four unit heaters **10***a*, **10***b*, **10***c*, and **10***d* may have a plurality of heating regions symmetrical to a virtual central vertical line or horizontal line passing through the center of the large-area heater **100**. For example, referring to the first unit heater **10***a* (same as the unit heater **10** of FIG. **1**), the second unit heater **10***b* is symmetrical with the first unit heater **10***a*, and the third unit heater **10***c* rotates the first unit heater **10***a* by 180 degrees. Instead of symmetrically arranging the unit heaters **10***a*, **10***b*, **10***c*, and **10***d*, the heater pattern may be directly disposed on an insulating plate having the same size as the heating plates of width 2L.sub.1 and length 2L.sub.2. However, it is difficult to maintain the arrangement of the heater pattern during the manufacturing process, and in particular, it is difficult to secure heating/pressurization equipment capable of processing a large-area insulating plate.
(19) The heating plate may include a central region **110** to which heat is concentrated and a peripheral region **120** surrounding the central region **110**. When the four unit heaters **10***a*, **10***b*, **10***c*, and **10***d* are symmetrically arranged, the first heating regions A**1**, A**2**, A**3**, and A**4** of each of the unit heaters **10***a*, **10***b*, **10***c*, and **10***d* heat the central region **110**, and the second heating regions B**1** to D**4** heat peripheral region **120**. In particular, the central region **110** is a region that directly transfers heat to an object to be heated, for example, an OLED substrate, and is a region requiring precise temperature uniformity. On the other hand, the peripheral region **120** may transmit heat directly to the object to be heated depending on the heating method (for example, whether there is physical contact between the heating plate and the object to be heated), but may mainly transfer heat to the central region **110** in order to reduce a temperature deviation of the central region **110**.
(20) In the large area heater **100**, the ratio of the first heating regions A**1**, A**2**, A**3**, and A**4** corresponding to the central region **110** to the entire area is relatively higher than that of the second heating regions B**1** to D**4** corresponding to the peripheral region **120**, while the amount of heat generated per area is relatively low compared to that of the peripheral region **120**. Since the

peripheral region **120** surrounds the central region **110**, heat generated in the peripheral region **120** tends to move to the central region **110**. Accordingly, the temperature of the central region **110** may be finely controlled through the peripheral region **120**. The heating regions B**1** to D**4** corresponding to the peripheral area **120** consist of independently power-controlled multi-zones B**1** to B**4** and single zones C**1** to D**4**, and the heat amount of each zone is different from each other. Similarly, the heating regions A**1** to A**4** corresponding to the central region **110** are configured as multi-zones, and the amount of heat generated in each zone is different from each other. By individually raising or lowering the temperature of each zone in the peripheral region **120** according to the temperature measurement in the central region **110**, the temperature deviation of the central region **110** can be maintained within a required range.
(21) FIG. **3** exemplarily illustrates coupling sub-patterns to form a heating region.
(22) Referring to FIG. **3**, the heater pattern **20** may be formed by coupling a plurality of sub-patterns **21** to **23**. The sub-patterns **21** to **23** may be, for example, bending patterns. The sub-patterns **21** to **23** may include a pair of connectors **21***a* and **21***b* and a sub-heating element **21***c* bent in a zigzag shape to have a plurality of bending points between the connectors. The sub-heating element **21***c* may be bent substantially at 90 degrees at the bending point.
(23) For example, the two sub-patterns **21** and **22** may be electrically coupled by fitting the connectors **21***a* and **22***a*. The connector **21***a* includes a slit **21***aa* extending from the bottom to the top, and the connector **22***a* includes a slit **22***aa* extending from the top to the bottom. For example, when fitted into the slits **21***aa* and **22***aa*, the left end of the connector **21***a* is located below the right end of the connector **22***a*, and the right end of the connector **21***a* is located above the left end of the connector **22***a*. Then, the connectors **21***a* and **22***a* may be coupled by welding or the like.
(24) Fitting using a slit is useful for adjusting the spacing between sub-patterns and positioning the sub-patterns. The extended length of the slit formed in the connector of the two sub-patterns to be coupled may be determined according to the spacing between the sub-patterns. For example, the length of each slit may be the same or different. In addition, the slit may be formed to be able to identify up and down or left and right. For example, the slit may extend obliquely, or one end of the slit may be formed to have a different left and right shape. On the other hand, the sub-patterns may be coupled by simply overlapping the connectors.
(25) FIG. **4** exemplarily illustrates a process for manufacturing a large-area heater.
(26) As described above, the large-area heater **100** may consist of a plurality of symmetrical unit heaters **10**. The unit heater **10** may be manufactured using a completed insulating substrate having a predetermined thickness or using a prepreg for manufacturing the insulating substrate.
(27) In step **200**, the insulating substrate is cut according to the size of the unit heater **10** to manufacture an upper insulating plate and a lower insulating plate. The insulating substrate has high electrical insulation. The insulating substrate may be a mica plate, but is not limited thereto. In addition to the mica plate, a metal place coated with a material having high electrical insulation and/or high thermal conductivity such as magnesium oxide, zirconium oxide, silicon oxide, titanium oxide, aluminum nitride, aluminum oxide, etc., non-metallic plate formed of ceramic, silicon carbide, etc., or synthetic resins such as polyimide, etc., may be used. Additionally, the upper insulating plate and/or the lower insulating plate may be coated with an anti-oxidation film. The anti-oxidation film can prevent the surface of the insulating plate from being oxidized.
(28) In step **210**, a plurality of sub-patterns are manufactured. The sub-pattern may be manufactured by, for example, etching or cutting a metal thin film formed of nickel, invar, sus, silver, tungsten, molybdenum, chromium, or an alloy thereof. Since the sub-pattern is made of a thin metal film, care must be taken in handling. To solve this problem, the metal thin film may be processed while being attached to the adhesive film.
(29) In step **220**, a high-temperature binder is applied to the surface to laminate the upper insulating plate and the lower insulating plate.
(30) In step **230**, the sub-patterns are attached to the surface of the lower insulating plate. The

heater pattern can be completed by welding the connectors of the attached sub-patterns. On the other hand, the sub-patterns may be welded in step **210** to attach the completed heater pattern to the surface of the lower insulating plate. Additionally, in order to protect the heater patterns and/or to increase electrical insulation, the heater patterns may be coated with an insulating film or the heater patterns may be interposed between the insulating films. The insulating layer or insulating film may be formed of a synthetic resin such as polyimide, silicone resin, epoxy resin, phenol resin, or the like.

(31) In step **240**, the lower insulating plate to which the heater pattern is attached is cured by a high-temperature and high-pressure press, and in step **250**, the upper insulating plate is attached to the lower insulating plate. In step **260**, the unit heater **10** consisting of the upper insulating plate, the heater pattern, and the lower insulating plate is cured by the high-temperature and high-pressure press.

(32) In step **270**, the plurality of unit heaters **10** are coupled to the heating plate to complete a large-area heater.

(33) FIG. **5** exemplarily illustrates one embodiment for implementing a different amount of heat generated for each heating region. Here, the heater pattern disposed in each zone is an example for helping understanding.

(34) Referring to FIG. **5**, the plurality of heater patterns **31** to **34** have different amounts of heat generated. To this end, a pattern density may be different for each heater pattern. The pattern density may be determined by the horizontal width p, the vertical width w, and the thickness t. The horizontal width p is the width of the heater pattern between two horizontally arranged bending points, the vertical width w is the width of the heater pattern between two vertically arranged bending points, and the thickness t is the line width of the heater pattern. The horizontal width p, the vertical width w, and the thickness t of the heater patterns **31** to **34** may be selected to implement a temperature range required by semiconductor equipment and to maintain a temperature deviation.

(35) Meanwhile, the heater pattern may be selected in consideration of the distance from the power connection region **11**. As the distance from the power connection region **11** increases, the power line **12** becomes longer and a resistance also increases, so that power loss may occur. Therefore, even when the same power is supplied to the heating region of the same area, a difference in the amounts of heat generated may occur depending on the length of the power line **12**. Therefore, in the case of the heating region far from the power connection region **11**, the resistance can be reduced by increasing the thickness t. On the other hand, in the case of the heating region close to the power connection region **11**, the resistance due to the power line **12** is small, and thus the thickness t may be reduced. For example, the pattern density of the heating region relatively far from the power connection region **11** may be smaller than the pattern density of the region relatively close to the power connection region **11**.

(36) A heating region relatively far from the power connection region **11**, for example, a**3**, may have a smaller amount of heat generated per unit area than an area relatively close to the power connection region **11**, for example b**3** or D. Accordingly, the temperature deviation required by the semiconductor manufacturing equipment can be achieved through temperature control of b**3** or D having a higher pattern density.

(37) FIG. **6** exemplarily illustrates another embodiment for implementing a different amounts of heat generated for each heating region.

(38) Referring to FIG. **6**, by varying the cross-sectional thickness of the heating elements **41**, **51**, and **61** for each heating region, the amount of heat generated may be different. When looking at the cross-section of the heater pattern, the thickness t**50** of the cross-section of the heating element **51** of the second heater pattern **50** is greater than the thickness t**40** of the cross-section of the heating element **41** of the first heater pattern **40**, but the third heater pattern **50** is smaller than the cross-sectional thickness of the heating element. Assuming that the pattern density and the distance from

the power connection region **11** are the same, the amount of heat generated by the heating elements **41**, **51**, and **61** may be different depending on the cross-sectional thickness.
(39) The aforementioned description for the present invention is exemplary, and those skilled in the art can understand that the invention can be modified in other forms without changing the technical concept or the essential feature of the invention. Therefore, it should be understood that the embodiments described above are illustrative and non-limiting in all respects. In particular, the features of the present invention described with reference to the drawings are not limited to the structures shown in the specific drawings, and may be implemented independently or in combination with other features.
(40) The scope of the invention is defined by the appended claims, not by the above detailed description, and it should be construed that all changes or modifications derived from the meanings and scope of the claims and equivalent concepts thereof are included in the scope of the invention.

Claims

1. A large-area heater comprising: a heating plate having a central region to which heat is concentrated and a peripheral region surrounding the central region; and a plurality of unit heaters configured for heating at least a portion of the central region and at least a portion of the peripheral region, wherein the plurality of unit heaters comprises a plurality of heating regions, each of the plurality of heating regions being configured for generating a different associated amount of heat compared to the other of the plurality of heating regions, a plurality of power lines configured for independently transmitting power to each of the plurality of heating regions, to cause each of the plurality of heating regions to generate the different associated amount of heat at the same time as the other of the plurality of heating regions, and a plurality of power connection terminal pairs disposed at one corner of the unit heater to supply power to the plurality of power lines, wherein the number of the plurality of the power connection terminal pairs corresponds to the number of the plurality of heating regions.
2. The large-area heater of claim 1, wherein the heating region comprises a single zone; and a multi-zone having a plurality of sub-zones generating different amounts of heat, wherein the plurality of sub-zones are supplied with power through one power line.
3. The large-area heater of claim 1, wherein the plurality of unit heaters comprises left, right and vertically symmetrical heating regions.
4. The large-area heater of claim 1, wherein the unit heater comprises a first heating region corresponding to the central region; and a second heating region corresponding to the peripheral region, wherein a pattern density of a heater pattern disposed in the first heating region is smaller than a pattern density of a heater pattern disposed in the second heating region.
5. The large-area heater of claim 1, wherein the unit heater comprises a first heating region corresponding to the central region; and a second heating region corresponding to the peripheral region, wherein a cross-sectional thickness of a heater pattern disposed in the first heating region is different from a cross-sectional thickness of a heater pattern disposed in the second heating region.
6. The large-area heater of claim 1, wherein the heating region comprises a heater pattern formed by electrically connecting a plurality of sub-patterns.
7. The large-area heater of claim 6, wherein the plurality of sub-patterns comprises a first connector and a second connector, both having slit formed thereon and a sub heating element bent in a zigzag shape to have a plurality of bending points between the first connector and the second connector.
8. The large-area heater of claim 1, wherein a temperature deviation of the central region is adjusted by controlling a temperature of the second heating region corresponding to the peripheral region.